Template Reference

A comprehensive guide to using the LaTeX template

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Headings

這是正文範例。Body text with both English and 中文混排。

Heading 2 Example

段落之間空一行就會自動產生間距（由 preamble.tex 的 ParSkip 控制）。

Heading 3 Example

第三層標題。

Text Formatting

粗體 Bold、斜體 *Italic*、等寬字 `Monospace`、底線 Underline、~~刪除線 Strikethrough~~。

粗斜體、**粗體紅字**。

紅色文字、深藍色文字、森林綠文字。

小字、正常、大字、更大。

Hyphens and dashes: well-known, pp. 1–10, 2020–2024, LaTeX — a powerful tool.

English quotes: “double quotes” and ‘single quotes’.

可點擊的連結文字。

Lists

Unordered List

- First item
- Second item with sub-items:
  - Sub-item A
  - Sub-item B
- Third item

Ordered List

1. Step one
2. Step two
3. Step three

Custom Lists

(a) First with letter label

(b) Second with letter label

3. This starts at 3
4. This is 4

Description List

Term A Definition of term A.
Term B Definition of term B.

Callout Boxes

Note

This is an informational note with default title.

Caution: Memory Leak

This is a warning with a custom title.

Tip

This is a tip with default title.

Critical Bug

This is a danger alert with a custom title.

Blockquote

> *The only way to do great work is to love what you do.*
> *— Steve Jobs*

Mathematics

Inline math: $E = mc^2$, $\sum_{i=1}^{n} i = \frac{n(n+1)}{2}$.

$$\int_{-\infty}^{\infty} e^{-x^2}\, dx = \sqrt{\pi} \tag{1}$$

Reference: eq. (1).

$$a^2 + b^2 = c^2$$

$$\begin{aligned} f(x) &= x^2 + 2x + 1 \\ &= (x+1)^2 \end{aligned} \tag{2}$$

Theorem 6.1. *For any triangle with sides a, b, and c, and the angle C opposite side c:*

$$c^2 = a^2 + b^2 - 2ab\cos C$$

Definition 6.1. *A **group** $(G, \cdot)$ is a set G with a binary operation $\cdot$ satisfying closure, associativity, identity, and invertibility.*

Lemma 6.1. *If f is continuous on $[a, b]$, then f is bounded on $[a, b]$.*

Code

Inline code: `print("hello")` is a Python statement.

```python
def fibonacci(n: int) -> int:
    """Return the n-th Fibonacci number."""
    if n ≤ 1:
        return n
    return fibonacci(n - 1) + fibonacci(n - 2)

print(fibonacci(10))  # 55
```

```bash
# shell script example
for i in $(seq 1 5); do
    echo "iteration $i"
done
```

Figures

Basic Usage

Reference: fig. 1 shows the full-width figure.

Size Control

Side-by-Side Figures

See fig. 4 and fig. 5.

Shadow Box

Tables

Table 1: A sample table with booktabs.

Name	Score	Grade
Alice	95	A+
Bob	87	A
Charlie	72	B

See table 1.

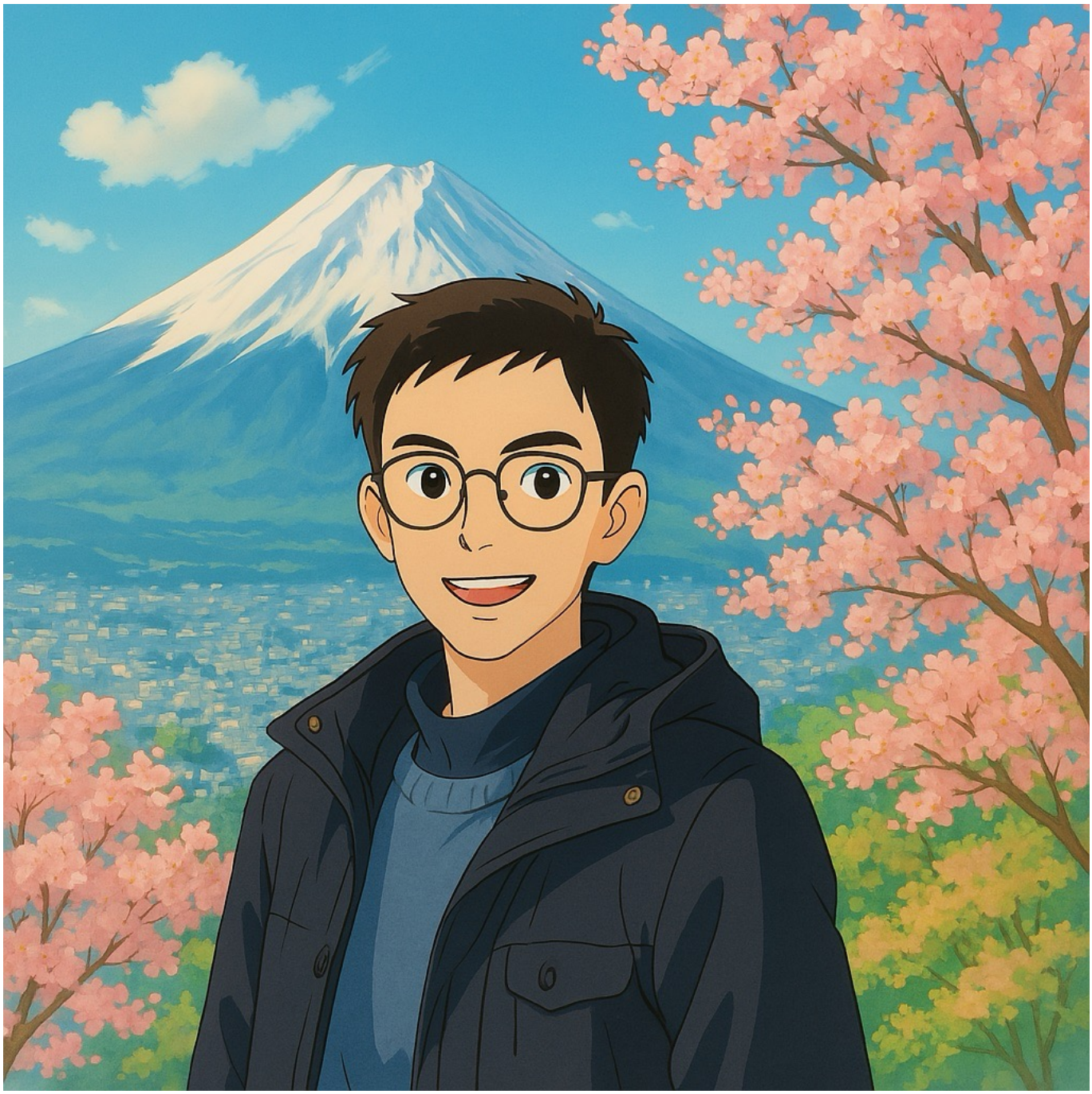

Figure 1: 範例圖片（預設滿版寬）。

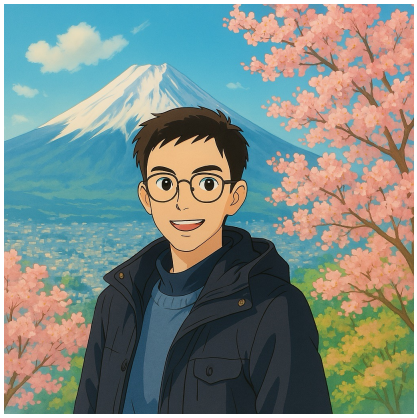

Figure 2: 縮到 30% 頁寬。

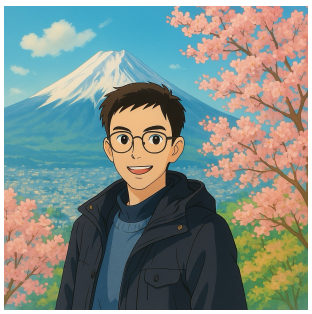

Figure 3: 限制高度 4cm。

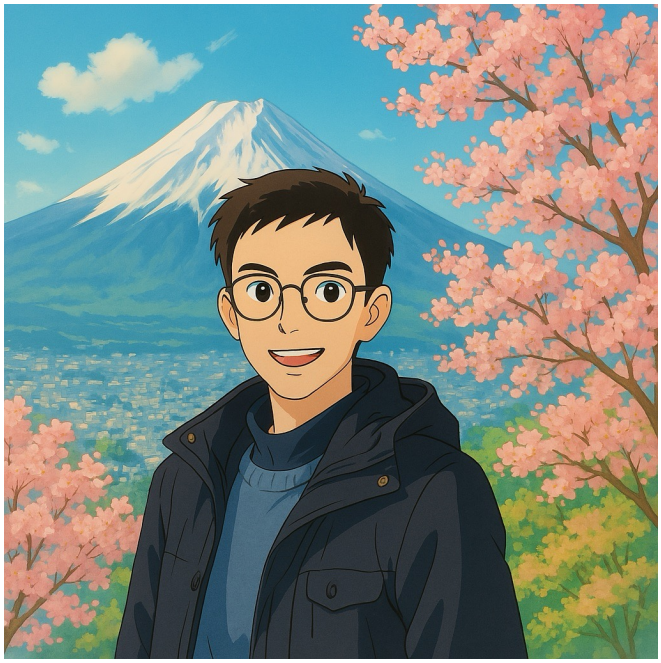

Figure 4: 左圖。

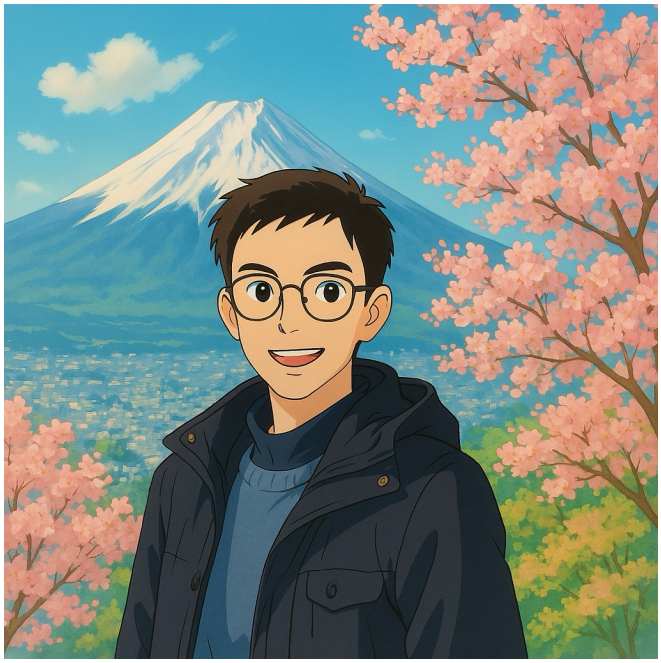

Figure 5: 右圖。

Figure 6: 圖片加上邊框與模糊陰影（`\shadowbox`）。

Algorithm

Algorithm 1: Binary Search

Input: Sorted array $A[1..n]$, target t

Output: Index of t in A, or -1

```
lo ← 1, hi ← n;
while lo ≤ hi do
    mid ← ⌊(lo + hi)/2⌋;
    if A[mid] = t then
        return mid;
    else if A[mid] < t then
        lo ← mid + 1;
    else
        hi ← mid − 1;
    end
end
return −1;
```

See algorithm 1.

Cross-References

See fig. 1, table 1, eq. (1), and algorithm 1.

Footnotes and Margin Notes

This sentence has a footnote[1].

This paragraph has a margin note.

Remem
this poi

Colors

Red, Blue, ForestGreen.

This entire paragraph is in BrickRed. The color applies until the closing brace.

Custom purple text. Also: 50% blue + 50% black.

highlighted text in a sentence.

Multi-Column Layout

Two Columns

[1] This is the footnote content, displayed at the bottom of the page.

Left column. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore.

Right column. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip.

Three Columns

Column 1. Short text.

Column 2. Medium length paragraph for demonstration.

Column 3. Another column.

Page Control

Todo Examples

This is an inline todo note for review.

Task list:

- ☐ Write the introduction
- ☐ Add figures
- ☒ ~~Set up template~~

Bibliography